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Breaking Report

Catalytic asymmetric synthesis, Palladium, Optically active aryl allenes, Optically active bis-allenes, Selectivities

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Highly Selective Nucleophilic 4-Aryl-2,3-allenylation of Malonates†

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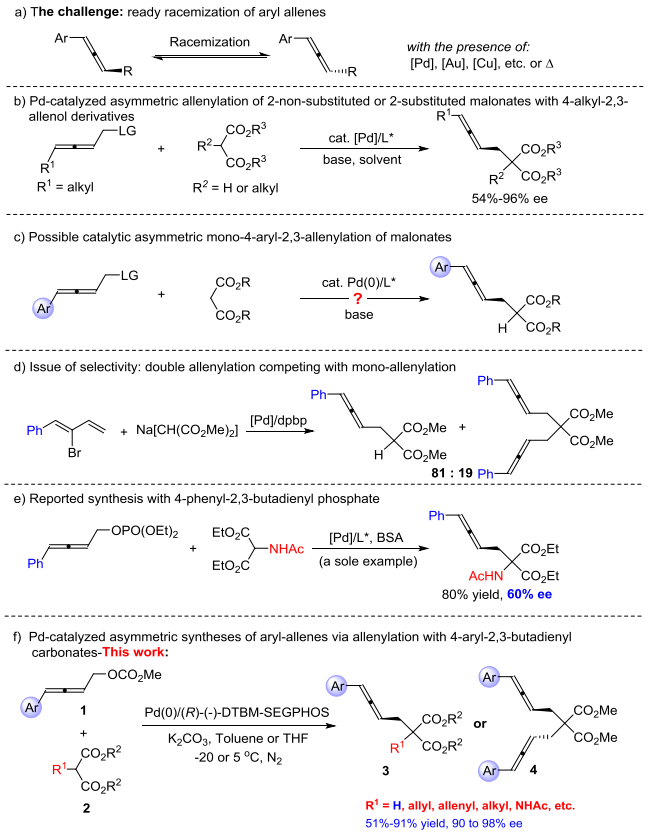
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Summary of main observation and conclusion Allenes are a class of very important compounds and the development of straightforward, efficient, and highly enantioselective synthetic strategies for allenes have attracted extensive interests. Along this line, it is well known that aryl-substituted allenes may be readily racemized, thus, difficult to prepare in high ee. Herein, an efficient palladium-catalyzed nucleophilic allenylation of malonates with racemic 4-aryl-2,3-butadienyl carbonates has been developed. The selectivity issue of mono- *vs*. bis-allenylation with 2-non-substituted malonates has been addressed. By utilizing (*R*)-(–)-DTBM-SEGPHOS (5,5'-bis[di(3,5-di-*t*-butyl-4-methoxyphenyl)phosphino]-4,4'-bi-1,3-benzodioxole) as a chiral ligand, various aryl-substituted allenes and bisallenes have been prepared with good to excellent yields with high chemoselectivity and enantioselectivity under mild reaction conditions. Au-catalyzed cycloisomerization and APK (allenic Pauson–Khand) reactions affording optically active mono- and bicyclic products have been demonstrated.

Background and Originality Content

Allene units are present widely in natural products and non-natural products with unique bioactivities.[1] Also, they have been demonstrated as versatile building blocks in organic syntheses,[2] providing novel methodologies to construct diverse and complex compounds, especially optically active chemicals via chirality transfer strategy.[3] Thus, design and development of new catalytic recipes to access functionalized chiral allenes are of high current interest.[4,5] However, it is well known that aryl-substituted allenes racemize easily in the presence of metal catalysts, which results mostly from the interaction of metal catalysts with the conjugate C=C in the allene parts (Scheme 1a)[6] and highly enantioselective syntheses of chiral aryl allenes are still challenging. Recently, catalytic asymmetric allenylation of malonates with 2,3-allenol derivatives[7] or 1,3-dien-2-yl derivatives[8] sharing the same intermediacy has been developed as an efficient approach for enantioselective syntheses of alkyl-substituted allenes (Scheme 1b). We wonder whether such a protocol may be applied to the syntheses of the thermo-sensitive aryl-allenes (Scheme 1c). The challenges are as follows: (1) such reactions with 4-aryl-2,3-allenol derivatives may be too reactive, resulting in the formation of a mixture of mono- and bis-allenylation products with non-substituted malonates (for such a report with 1,3-dien-2-yl derivatives, see: Scheme 1d);[9] (2) the reported enantioselectivity with 4-phenyl-2,3-allenyl derivatives is low even with the sterically bulky diethyl 2-acetamidomalonate (Scheme 1e);[7a,8] (3) a more active catalytic system, which will work at very mild reaction conditions to avoid the racemization of *in situ* generated sensitive aryl-substituted allenes, is required. Herein, we wish to report our recent realization of Pd-catalyzed highly chemoselective mono-allenylation of non-substituted malonates (Scheme 1f). The protocol also works well with allenylation of sterically bulkier 2-substituted malonates and bisallenylation of 2-non-substituted malonates under mild reaction conditions.

Scheme 1 Ready racemization of chiral arylallenes and syntheses of allenes *via* nucleophilic allenylation of different malonates



Results and Discussion

We commenced our study on the reaction of allenylic carbonate **1a** with the most challenging dimethyl malonate **2a** to address the selectivity issue of monoallenylation over bisallenylation. As expected, we found that treatment of **1a** with **2a** catalyzed by $Pd_2(dba)_3{\cdot}CHCl_3$ with (±)-BINAP as ligand and K_2CO_3 as base in THF (tetrahydrofuran) furnished (±)-**3aa** smoothly in 68% yield together with the bisallenylation product (±)-**4aa** (entry 1, Table 1) in a poor selectivity of 4 : 1.[9] Increasing the loading of **2a** would slightly increase the selectivity of monoallenylation product

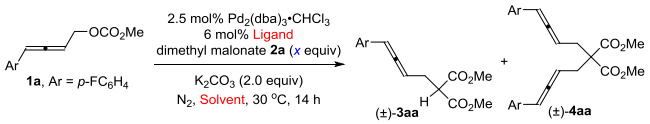
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† Dedicated to the 30th Anniversary of State Key Laboratory of Organometallic Chemistry.

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Table 1 Identifying the optimal reaction conditions for the reaction of **1a** with **2a**[a]



Entry	Ligand	*x*	Solvent	(±)-**3aa**[b] /%	(±)-**3aa**:(±)-**4aa**[b]	Recovery of **1a**[b]
1	(±)-BINAP	2.0	THF	68	81:19	—[c]
2	(±)-BINAP	3.0	THF	71	91:9	—[c]
3	(±)-BINAP	4.0	THF	63	93:7	—[c]
4	DPPE	4.0	THF	6	100:0	73
5	PPh_3[d]	4.0	THF	—[c]	—[c]	61
6	(±)-BINAP	4.0	Dioxane	68	100:0	—[c]
7	(±)-BINAP	4.0	Toluene	74	100:0	—[c]
8	**(±)-BINAP**	**2.0**	**Toluene**	**80 (78)**[e]	**100:0**	—[c]

[a] The reaction of $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%), ligand (0.012 mmol), K_2CO_3 (0.4 mmol), **2a** (*x* equiv)/solvent (1.0 mL), and **1a** (0.2 mmol) in solvent (1.0 mL) was stirred at 30 °C. [b] Data determined by NMR analysis of the crude reaction mixture. [c] Not detected. [d] 12 mol% PPh_3 was used. [e] Isolated yield.

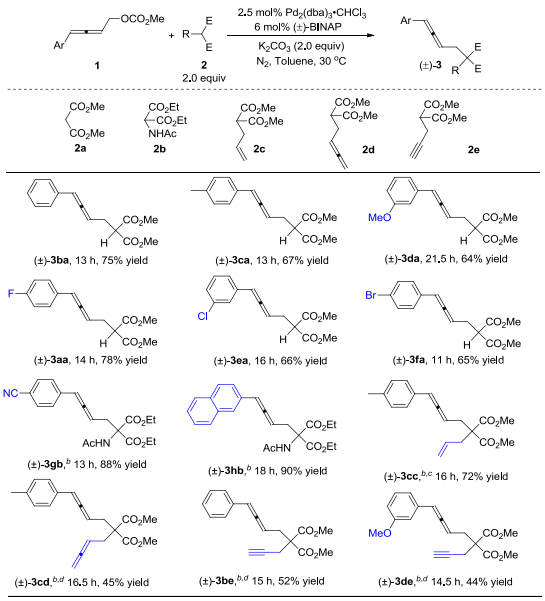
(±)-**3aa** over bisallenylation product (±)-**4aa** (entries 2 and 3, Table 1). The screening of ligands inferred that (±)-BINAP (2,2'-bis(diphenylphosphino)-1,1'-binaphthalene) was still the best choice (entries 4—5, Table 1). Interestingly, a profound solvent effect was observed: By using dioxane as solvent, the reaction afforded monoallenylation product (±)-**3aa** in 68% yield, exclusively (entry 6, Table 1). A much higher yield with an exclusive chemoselectivity was observed using toluene as solvent (entry 7, Table 1). In this solvent, reducing the amount of **2a** further increased the yield of (±)-**3aa** to 80%, with no formation of bisallenylation product (±)-**4aa**.

Next, the substrate scope for the synthesis of racemic 1,3-disubstituted arylallenes was investigated (Table 2). The reaction of dimethyl malonate (**2a**) with allenylic carbonates **1a**—**1f** afforded the monoallenylation products (±)-**3aa** to (±)-**3fa** exclusively in 64%—78% yields. As for nucleophilic reagents **2b**—**2e** equipped with useful functional groups, such as 2-acetamide, 2-allyl, 2-allenyl, and 2-propargyl, the reaction with different 4-aryl-2,3-allenylic carbonates **1** containing electron-withdrawing and electron-donating groups on the aryl ring could all deliver the desired products with moderate to excellent yields (44%—90%), regardless of the location of the substituents.

We went further to develop the asymmetric version of this transformation by using 4-aryl-2,3-allenylic carbonate **1a** and the much less sterically hindered dimethyl malonate (**2a**) as the model substrates. Under the catalysis of $Pd_2(dba)_3{\cdot}CHCl_3$, different chiral ligands were tried (entries 1—6, Table 3) and (*R*)-(–)-DTBM-SEGPHOS was found to be the preferred ligand. The results in Table 3 showed that both the loadings of **2a** and base had an obvious effect on the enantioselectivity of (R_a)-**3aa** and the selectivity of monoallenylation product (R_a)-**3aa**/bisallenylation product (R_a,R_a)-**4aa** (entries 6—9, Table 3). With 2.0 equiv each of **2a** and K_2CO_3, the reaction afforded (R_a)-**3aa** in 77% yield with an ee of 93% and the ratio of (R_a)-**3aa**/(R_a,R_a)-**4aa** was improved to 92/8 in THF (entry 7, Table 3). Replacing THF with toluene, the reaction furnished monoallenylation product (R_a)-**3aa** overwhelmingly with an ee of 87% (entry 8, Table 3). Further optimization led to the establishment of the optimal reaction conditions for further scope study: with 2.5 equiv of **2a**, the reaction in toluene at –20 °C yielded (R_a)-**3aa** exclusively with an ee of 91% (entry 11, Table 3).

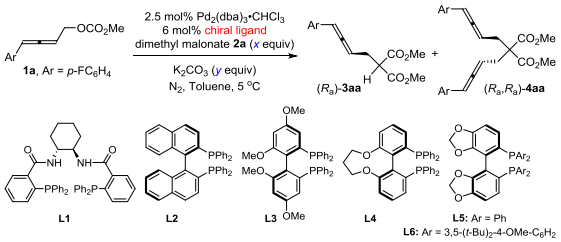
With the optimal reaction conditions in hand (entry 11, Table 3), the substrate scope of 2,3-allenylic carbonates with 2-non-sub-

Table 2 The synthesis of racemic 1,3-disubstituted arylallenyl malonates[a]



[a] The mixture of $Pd_2(dba)_3{\cdot}CHCl_3$ (0.005 mmol), (±)-BINAP (0.012 mmol), K_2CO_3 (0.4 mmol), **2** (0.4 mmol)/toluene (1.0 mL) and **1** (0.2 mmol)/toluene (1.0 mL) was added and the resulting mixture was stirred at 30 °C. The yields were gained after column chromatographic separation on silica gel. [b] THF was used instead of toluene. [c] DTBM-SEGPHOS (5,5'-bis[di(3,5-di-*t*-butyl-4-methoxyphenyl)phosphino]-4,4'-bi-1,3-benzodioxole) was used instead of (±)-BINAP. [d] 5 mol% Pd(dmdba)$_2$ was used instead of 2.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$.

Table 3 Identifying the best ligand and optimization of reaction conditions[a]



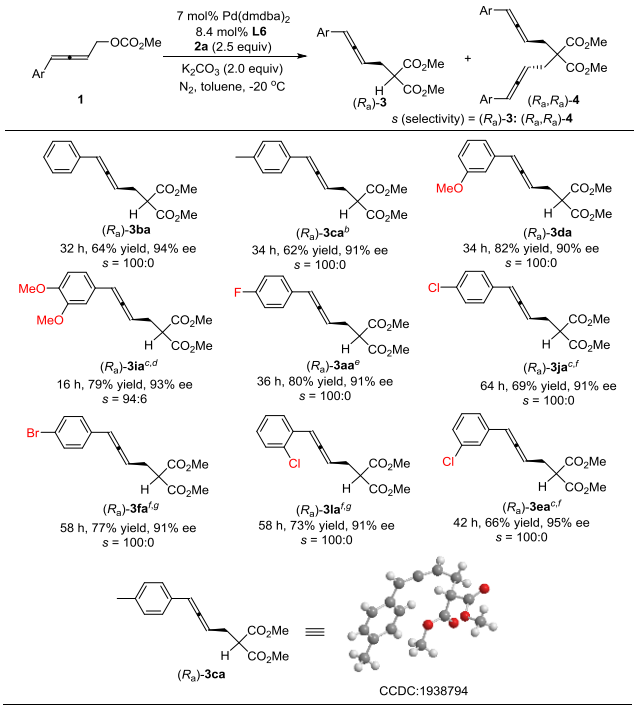
Entry	Ligand	*x*/*y*	*t*/h	Yield of (R_a)-**3aa**[b]/%	ee of (R_a)-**3aa**[c]/%	(R_a)-**3aa**: (R_a,R_a)-**4aa**[d]
1[e]	**L1**	2.0/2.2	10	—[f]	—[f]	—
2[e]	**L2**	2.0/2.2	10	77	40	89:11
3[e]	**L3**	2.0/2.2	11	74	75	84:16
4[e]	**L4**	2.0/2.2	13	73	84	82:18
5[e]	**L5**	2.0/2.2	11.5	74	77	89:11
6[e]	**L6**	2.0/2.2	18	69	84	85:15
7[e]	**L6**	2.0/2.0	12	77	93	92:8
8	**L6**	2.0/2.0	13	80	86	97:3
9[g]	**L6**	2.5/2.0	17	67	87	100:0
10[h,i]	**L6**	2.5/2.0	21	73	89	100:0
11[i,j]	**L6**	**2.5/2.0**	**36**	**80**	**91**	**100:0**

[a] A mixture of $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%), chiral ligand (0.012 mmol), K_2CO_3 (*y* equiv), and dimethyl malonate **2a** (*x* equiv)/toluene (1.5 mL) was stirred at 25 °C for 30 min. The mixture was stirred at 5 °C for 10 min and then **1a** (0.2 mmol)/toluene (0.5 mL) was added. [b] Isolated yields. [c] The ees of (R_a)-**3aa** were determined by chiral HPLC analysis. [d] Data determined by NMR analysis of the crude product. [e] THF was used as solvent instead of toluene. [f] (R_a)-**3aa** was not afforded and 89% of **1a** was recovered. [g] The reaction was conducted at –5 °C. [h] The reaction was conducted at –15 °C. [i] 5 mol% Pd(dmdba)$_2$ was used instead of 2.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$. [j] The reaction was conducted at –20 °C.

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Table 4 The substrate scope of allenylic carbonates with dimethyl malonate[a]

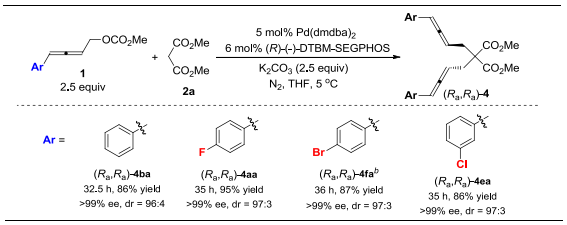


[a] The mixture of 7 mol% $Pd(dmdba)_2$, 8.4 mol% **L6**, K_2CO_3 (0.4 mmol), and **2a** (0.5 mmol)/solvent (1.5 mL) was stirred at 25 °C for 30 min. The mixture was stirred at –20 °C for 10 min and then **1** (0.2 mmol)/solvent (0.5 mL) was added. The *s* values were calculated by NMR yields of (R_a)-**3** and (R_a,R_a)-**4** as determined by NMR analysis of the crude product. [b] 5 mol% PPh_3 and 9.8 mol% **L6** were used as ligands. [c] 2.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$ and 6 mol% **L6** were used. [d] The reaction using 2.0 equiv **2a** with THF as solvent was conducted at 5 °C. [e] 5 mol% $Pd(dmdba)_2$ and 6 mol% **L6** were used. [f] The mixed solvent (Toluene : THF = 30 : 1) was used. [g] 3.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$ was used.

stituted dimethyl malonate **2a** was studied (Table 4). Generally speaking, electronic and steric effects have a very limited impact on this reaction and the desired products were afforded in good yields (62%—82%) with high ees (90%—95%) and selectivity (*s* = 94 : 6 to 100 : 0). Substituents on phenyl rings in allene **1**, such as F, Cl, Br, and OMe, are all compatible. Importantly, the mono-substituted malonate-type products (R_a)-**3aa** to (R_a)-**3la** still possess a handle for further synthetic elaboration. The absolute configuration of major enantiomer of **3ca** was determined by X-ray single crystal diffraction study to be *R*.

Moreover, bis(arylallenylation) of dimethyl malonate **2a** was also realized in good to excellent yields (86%—95%) and excellent enantio- and diastereoselectivities (>99% ee, 96 : 4 to 97 : 3 dr) upon the molar ratio of **1** : **2a** being adjusted to 2.5 : 1 (Table 5).

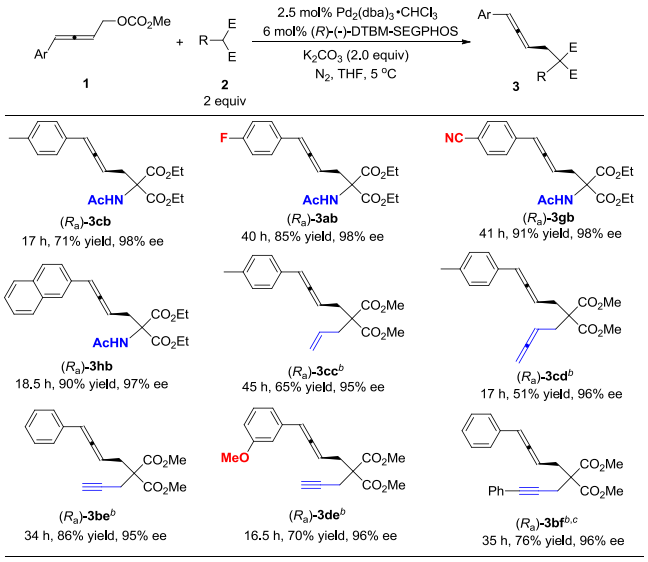
Table 5 Highly enantio- and diastereoselective bis(arylallenylation)[a]



[a] The mixture of $Pd(dmdba)_2$ (0.01 mmol), (*R*)-(–)-DTBM-SEGPHOS (0.012 mmol), K_2CO_3 (0.5 mmol), and **2a** (0.2 mmol)/THF (1.5 mL) was stirred at 25 °C for 30 min. The mixture was stirred at 5 °C for 10 min and then **1** (0.5 mmol)/THF (0.5 mL) was added. The ee values and drs were determined by chiral HPLC analysis. [b] 3.0 equiv **1f** was used.

Then the scope of 2-substituted malonates for this enantioselecctive transformation was evaluated (Table 6). Compared to **2a**, 2-substituted malonates (**2b**—**2f**) could afford the arylallenes with relatively higher ees as expected. When diethyl 2-acetamidomalonate (**2b**) was exploited as nucleophile, allenylic carbonates with the aryl group substituted with 4-Me, 4-F, 4-CN, and 2-naphthyl gave the products with 97%—98% ee.[7a,8] In addition, 2-allyl, 2-allenyl, 2-propargyl, and 2-(3-phenylpropargyl)-substituted malonates all worked smoothly. These synthetically useful groups combined with the chiral allene unit could afford other chiral cyclic molecules *via* different types of [4+2],[3f,3g] [2+2+1][3h] or [2+2][10] cycloaddition reactions.

Table 6 The substrate scope of allenylic carbonates with substituted malonates[a]



[a] The mixture of $Pd_2(dba)_3{\cdot}CHCl_3$ (0.0125 mmol), (*R*)-(–)-DTBM-SEGPHOS (0.03 mmol), K_2CO_3 (1.0 mmol), and **2** (1.0 mmol)/THF (4.0 mL) was stirred at 25 °C for 30 min. The mixture was stirred at 5 °C for 10 min and then **1** (0.5 mmol)/THF (1.0 mL) was added. [b] 5 mol% $Pd(dmdba)_2$ was used instead of 2.5 mol% $Pd_2(dba)_3{\cdot}CHCl_3$. [c] **2f** was dimethyl 2-(3-phenylprop-2-ynyl) malonate.

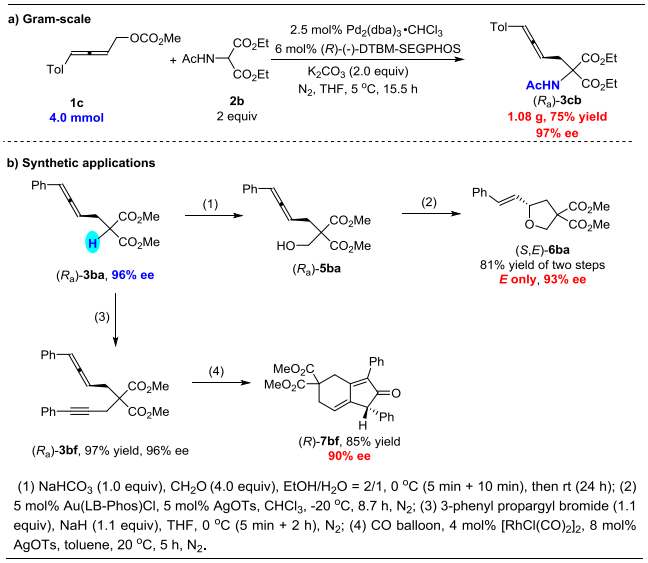
This transformation could be easily conducted on a gram-scale: The reaction of allenylic carbonate **1c** (4.0 mmol) with diethyl 2-acetamidomalonate (**2b**) gave (R_a)-**3cb** (1.08 g) in 75% yield and 97% ee (Scheme 2). Synthetic applications have been demonstrated to highlight the potential values of this methodology. Treatment of (R_a)-**3ba** with HCHO (37%, aq.) in the presence of $NaHCO_3$ gave γ-allenol (R_a)-**5ba**, which subsequently produced optically active tetrahydrofuran derivative (*S*,*E*)-**6ba** with an exclusive *E*-C=C bond through gold-catalyzed cycloisomerization.[3e] Reaction of (R_a)-**3ba** with 3-phenylpropargyl bromide gave (R_a)-**3bf** and its subsequent APK (allenic Pauson–Khand) reaction of (R_a)-**3bf** in the presence of CO co-catalyzed by $[Rh(CO)_2Cl]_2$/AgOTs afforded the optically active bicyclopentenone compound (*R*)-**7bf** with a highly sensitive chiral center *via* chirality transfer strategy.[3h]

Finally, a rationale was proposed (Scheme 3): S_N2'-type oxidation addition of PdL* with (R_a)-**1** or (S_a)-**1** from the back side of the C-OCO_2Me bond would generate the same intermediate *E*-η^1-1,3-dienyl Pd,[11] which would immediately undergo delocalization to yield intermediate η^3-**Int A**. Subsequently, the nucleophile would attack from the back side of the terminal carbon atom to generate the chiral allene **3** in *R* configuration, as shown in the model on the up right corner.

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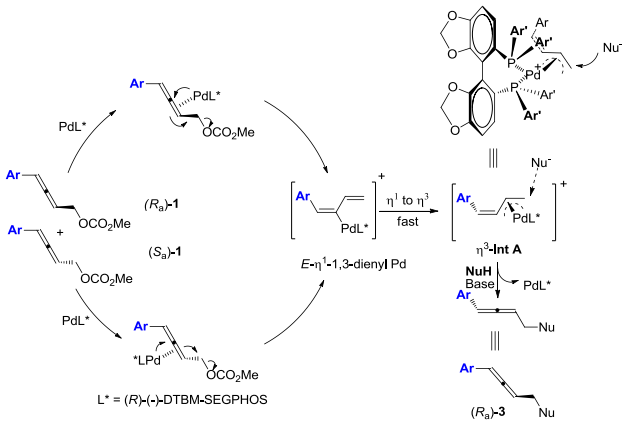
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Scheme 2 Gram-scale reaction and synthetic applications



(1) $NaHCO_3$ (1.0 equiv), CH_2O (4.0 equiv), EtOH/H_2O = 2/1, 0 °C (5 min + 10 min), then rt (24 h); (2) 5 mol% Au(LB-Phos)Cl, 5 mol% AgOTs, $CHCl_3$, -20 °C, 8.7 h, N_2; (3) 3-phenyl propargyl bromide (1.1 equiv), NaH (1.1 equiv), THF, 0 °C (5 min + 2 h), N_2; (4) CO balloon, 4 mol% $[RhCl(CO)_2]_2$, 8 mol% AgOTs, toluene, 20 °C, 5 h, N_2.

Scheme 3 A rationale for the formation of product (R_a)-**3**



Conclusions

In summary, we have developed a Pd(0)-catalyzed synthesis of the easily racemizable aryl-substituted allenes *via* highly selective nucleophilic allenylation of 2-non-substituted and 2-substituted malonates with allenylic carbonates. The highly enantioselective version of this transformation has been successfully realized with (*R*)-(–)-DTBM-SEGPHOS as the chiral ligand. This method features high efficiency, high selectivity, good substrate scope, and very mild reaction conditions avoiding racemization. Synthetic applications have also been demonstrated. We are actively pursuing other targets in this area.

Experimental

The asymmetric synthesis of (R_a)-**3aa** (Ssh-07-128): To a dry Schlenk tube were added K_2CO_3 (55.4 mg, 0.4 mmol), (*R*)-(–)-DTBM-SEGPHOS (14.2 mg, 0.012 mmol) inside a glove box. Then $Pd(dmdba)_2$ (8.3 mg, 0.01 mmol), and **2a** (65.7 mg, 0.5 mmol)/toluene (1.5 mL) were added into this Schlenk tube under nitrogen atmosphere outside the glove box. After being stirred at 25 °C for 30 min, the resulting mixture was stirred at –20 °C for 10 min. Then **1a** (43.3 mg, 0.2 mmol)/toluene (0.5 mL) was added with stirring. After being stirred for 36 h at –20 °C, the reaction was completed as monitored by TLC. Filtration through a short column of silica gel (eluent: ethyl acetate (10 mL × 3)) and evaporation afforded a crude product (R_a)-**3aa** exclusively ((R_a)-**3aa** : (R_a,R_a)-**4aa** = 100 : 0, which was calculated by NMR yields of (R_a)-**3aa** and (R_a,R_a)-**4aa** as determined by ^{1}H NMR analysis of the crude product). Column chromatography on silica gel afforded (R_a)-**3aa** (44.8 mg, 80%) (eluent: petroleum ether (60—90 °C)/ethyl acetate/DCM = 20/1/1) as a liquid: 91% ee (HPLC conditions; Chiralcel OD-H column, *n*-hexane/*i*-PrOH = 90/10, 0.3 mL/min, λ = 214 nm, t_R (major) = 22.7 min, t_R (minor) = 23.7 min); $[\alpha]_D^{20}$ = –197.1 (*c* = 0.63, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.26—7.17 (m, 2H, ArH), 7.03—6.93 (m, 2H, ArH), 6.22—6.12 (m, 1H, =CH), 5.62 (q, *J* = 6.0 Hz, 1H, =CH), 3.73 (s, 3H, OCH_3), 3.64—3.52 (m, 4H, CH + OCH_3), 2.85—2.62 (m, 2H, CH_2); ^{13}C NMR (75 MHz, $CDCl_3$) δ: 204.9 (d, *J* = 2.0 Hz), 169.2, 169.0, 162.0 (d, *J* = 244.9 Hz), 130.0 (d, *J* = 3.1 Hz), 128.2 (d, *J* = 7.4 Hz), 115.5 (d, *J* = 21.3 Hz), 95.5, 92.0, 52.6, 52.5, 50.9, 27.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ: –115.6 (s, 1 F); IR (neat) *v*: 3002, 2955, 2847, 1951, 1753, 1737, 1602, 1508, 1436, 1342, 1226, 1157, 1094, 1039 cm^{-1}; MS (EI, 70 eV) *m/z* (%): 279 (M^+ + 1, 5.38), 278 (M^+, 31.41), 146 (100); HRMS calcd. for $C_{15}H_{15}O_4F$ [M^+]: 278.0954, found: 278.0952.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202000300.

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References

[1] (a) Kavanagh, F.; Hervey, A.; Robbins, W. J. Antibiotic substances from basidiomycetes V. Poria corticola, poria tenuis and unidentified basidiomycete. *Proc. Natl. Acad. Sci. U. S. A.* **1950**, *36*, 1–7; (b) Horler, D. F. (–) Methyl *n*-tetradeca-trans-2,4,5-trienoate, an allenic ester produced by the male dried bean beetle, acanthoscelides obtectus (Say). *J. Chem. Soc.* (*C*) **1970**, 859–862; (c) Cox, P. J.; Imre, S.; Islimyeli, S.; Thomson, R. H. Obusallene I, a new halogenated allene from laurencia obtusa. *Tetrahedron Lett.* **1982**, *5*, 579–580; (d) Gorins, G.; Kuhnert, L.; Johnson, C. R.; Marnett, L. J. (Carboxyalkyl)benzy propargyl ethers as selective inhibitors of leukocyte-type 12-lipoxygenases. *J. Med. Chem.* **1996**, *39*, 4871–4878; (e) McGrath, M. J.; Fletcher, M. T.; König, W. A.; Moore, C. J.; Cribb, B. W.; Allsopp, P. G.; Kitching, W. A suite of novel allenes from australian melolonthine scarab beetles. Structure, synthesis, and stereochemistry. *J. Org. Chem.* **2003**, *68*, 3739–3748; (f) Hoffmann-Röder, A.; Krause, N. Synthesis and properties of allenic natural products and pharmaceuticals. *Angew. Chem. Int. Ed.* **2004**, *43*, 1196–1216; (g) Jian, Y. J.; Wu, Y. The enantioselective total synthesis of nemotin. *Org. Biomol. Chem.* **2010**, *8*, 811–821.

[2] (a) Hiroi, K.; Kato, F.; Yamagata, A. Asymmetric direct α,β-functionalization of allenes *via* asymmetric carbopalladation. *Chem. Lett.* **1998**, 397–398; (b) Ogasawara, M.; Nagano, T.; Hayashi, T. A new route to methyl (*R*,*E*)-(–)-Tetradeca-2,4,5-trienoate (Pheromone of Acanthoscelides obtectus) utilizing a Palladium-catalyzed asymmetric allene formation reaction. *J. Org. Chem.* **2005**, *70*, 5764–5767; (c) Crouch, I. T.; Neff, R. K.; Frantz, D. E. Pd-catalyzed asymmetric β-Hydride elimination en route to chiral allenes. *J. Am. Chem. Soc.* **2013**, *135*, 4970–4973; (d) Tang, X.; Huang, X.; Cao, T.; Han, Y.; Jiang, X.; Lin, W.; Tang, Y.; Zhang, J.; Yu, Q.; Fu, C.; Ma, S. $CuBr_2$-catalyzed enantioselective routes to highly functionalized and naturally occurring allenes. *Org. Chem. Front.* **2015**, *2*, 688–691; (e) Jiang, X.; Xue, Y.;

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16147065, 2020, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.202000300 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ma, S. Aerobic oxidation and EATA-based highly enantioselective synthesis of lamenallenic acid. *Org. Chem. Front.* **2017**, *4,* 951–957; (f) Oonishi, Y.; Hosotani, A.; Yokoe, T.; Sato, Y. Rhodium(I)-catalyzed enantioselective hydroacylation of racemic allenals *via* dynamic kinetic resolution. *Org. Lett.* **2019**, *21*, 4120–4123.

[3] (a) Campolo, D.; Gastaldi, S.; Roussel, C.; Bertrand, M. P.; Nechab, M. Axial-to-central chirality transfer in cyclization processes. *Chem. Soc. Rev.* **2013**, *42*, 8434–8466; (b) Neff, R. K.; Frantz, D. E. Recent applications of chiral allenes in axial-to-central chirality transfer reactions. *Tetrahedron* **2015**, *71*, 7–18; (c) Alonso, J. M.; Quirós, M. T.; Muñoz, M. P. Chirality transfer in metal-catalysed intermolecular addition reactions involving allenes. *Org. Chem. Front.* **2016**, *3*, 1186–1204; (d) Amako, Y.; Arai, S.; Nishida, A. Transfer of axial chirality through the nickel-catalysed hydrocyanation of chiral allenes. *Org. Biomol. Chem.* **2017**, *15*, 1612–1617; (e) Zhou, J.; Fu, C.; Ma, S. Gold-catalyzed stereoselective cycloisomerization of allenoic acids for two types of common natural γ-butyrolactones. *Nat. Commun.* **2018**, *9*, 1654–1663; (f) Han, Y.; Ma, S. Rhodium-catalyzed highly diastereoselective intramolecular [4+2] cycloaddition of 1,3-disubstituted allene-1,3-dienes. *Org. Chem. Front.* **2018**, *5*, 2680–2684; (g) Han, Y.; Qin, A.; Ma, S. One stone for three birds-rhodium-catalyzed highly diastereoselective intramolecular [4+2] cycloaddition of optically active allene-1,3-dienes. *Chin. J. Chem.* **2019**, *37*, 486–496; (h) Han, Y.; Zhao, Y.; Ma, S. Rhodium-catalyzed Pauson-Khand-type cyclization of 1,5-allene-alkynes: a chirality transfer strategy for optically active bicyclic ketones. *Chem.-Eur. J.* **2019**, *25*, 9529–9533.

[4] For selected reviews of the synthesis of allenes: (a) Krause, N.; Hoffmann-Röder, A. Synthesis of allenes with organometallic reagents. *Tetrahedron* **2004**, *60*, 11671–11694; (b) Brummond, K. M.; DeForrest, J. E. Synthesizing allenes today (1982—2006). *Synthesis* **2007**, *6*, 795–818; (c) Ogasawara, M. Catalytic enantioselective synthesis of axially chiral allenes. *Tetrahedron-Asymmetry* **2009**, *20*, 259–271; (d) Yu, S.; Ma, S. How easy are the syntheses of allenes? *Chem. Commun.* **2011**, *47*, 5384–5418; (e) Neff, R. K.; Frants, D. E. Recent advances in the catalytic syntheses of allenes: a critical assessment. *ACS Catal.* **2014**, *4*, 519–528; (f) Ye, J.; Ma, S. Conquering three-carbon axial chirality of allenes. *Org. Chem. Front.* **2014**, *1*, 1210–1224; (g) Chu, W.; Zhang, Y.; Wang, J. Recent advances in catalytic asymmetric synthesis of allenes. *Catal. Sci. Technol.* **2017**, *7*, 4570–4579.

[5] For recent reports on catalytic enantioselective syntheses of allenes, see: (a) Li, C.; Wang, X.; Sun, X.; Tang, Y.; Zheng, J.; Xu, Z.; Zhou, Y.; Dai, L. Iron porphyrin-catalyzed olefination of ketenes with diazoacetate for the enantioselective synthesis of allenes. *J. Am. Chem. Soc.* **2007**, *129*, 1494–1495; (b) Nishimura, T.; Makino, H.; Nagaosa, M.; Hayashi, T. Rhodium-catalyzed enantioselective 1,6-addition of arylboronic acids to enynamides: asymmetric synthesis of axially chiral allenylsilanes. *J. Am. Chem. Soc.* **2010**, *132*, 12865–12867; (c) Wan, B.; Ma, S. Enantioselective decarboxylative amination: synthesis of axially chiral allenyl amines. *Angew. Chem. Int. Ed.* **2013**, *52*, 441–445; (d) Wang, Y.; Zhang, W.; Ma, S. A room-temperature catalytic asymmetric synthesis of allenes with ECNU-Phos. *J. Am. Chem. Soc.* **2013**, *135*, 11517–11520; (e) Wang, M.; Liu, Z.; Zhang, X.; Tian, P.; Xu, Y.; Loh, T. P. Synthesis of highly substituted racemic and enantioenriched allenylsilanes *via* Copper-catalyzed hydrosilylation of (*Z*)-2-alken-4-ynoates with silylboronate. *J. Am. Chem. Soc.* **2015**, *137*, 14830–14833; (f) Yao, Q.; Liao, Y.; Lin, L.; Lin, X.; Ji, J.; Liu, X.; Feng, X. Efficient synthesis of chiral trisubstituted 1,2-allenyl ketones by catalytic asymmetric conjugate addition of malonic esters to enynes. *Angew. Chem. Int. Ed.* **2016**, *55*, 1859–1863; (g) Liu, Y.; Liu, X.; Hu, H.; Guo, J.; Xia, Y.; Lin, L.; Feng, X. Synergistic kinetic resolution and asymmetric propargyl claisen rearrangement for the synthesis of chiral allenes. *Angew. Chem. Int. Ed.* **2016**, *55*, 4054–4058; (h) Chu, W. D.; Zhang, L.; Zhou, Q.; Mo, F.; Zhang, Y.; Wang, J. Enantioselective Synthesis of trisubstituted allenes *via* Cu(I)-catalyzed coupling of diazoalkanes with terminal alkynes. *J. Am. Chem. Soc.* **2016**, *138*, 14558–14561; (i) Poh, J. S.; Makai, S.; Keutz, T. v.; Tran, D. N. Battilocchio, C.; Pasau, P.; Ley, S. V. Rapid asymmetric synthesis of disubstituted allenes by coupling of flow-generated diazo compounds and propargylated amines. *Angew. Chem. Int. Ed.* **2017**, *56*, 1864–1868; (j) Wei, X.-F.; Wakaki, T.; Itoh, T.; Li, H.-L.; Yoshimura, T.; Miyazaki, A.; Oisaki, K.; Hatanaka, M.; Shimizu, Y.; Kanai, M. Catalytic regio- and enantioselective proton migration from skipped enynes to allenes. *Chem* **2018**, *5*, 585–599; (k) Hölzl-Hobmeier, A.; Bauer, A.; Silva, A. V.; Huber, S. M.; Bannwarth, C.; Bach, T. Catalytic deracemization of chiral allenes by sensitized excitation with visible light. *Nature* **2018**, *564*, 240–243; (l) Poulsen, P. H.; Li, Y.; Lauridsen, V. H.; Jørgensen, D. K. B.; Palazzo, T. A.; Meazza, M.; Jørgensen, K. A. Organocatalytic formation of chiral trisubstituted allenes and chiral furan derivatives. *Angew. Chem. Int. Ed.* **2018**, *57*, 10661–10665; (m) Petrone, D. A.; Isomura, M.; Franzoni, I.; Rössler, S. L.; Carreira, E. M. Allenylic carbonates in enantioselective Iridium-catalyzed alkylations. *J. Am. Chem. Soc.* **2018**, *140*, 4697–4704; (n) Trost, B. M.; Zell, D.; Hohn, C.; Mata, G.; Maruniak, A. Enantio- and diastereoselective synthesis of chiral allenes by Palladium-catalyzed asymmetric [3+2] cycloaddition reactions. *Angew. Chem. Int. Ed.* **2018**, *57*, 12916–12920; (o) Huang, Y.; Pozo, J.; Torker, S.; Hoveyda, A. H. Enantioselective synthesis of trisubstituted allenyl-B(pin) compounds by phosphine-Cu-catalyzed 1,3-Enyne hydroboration. Insights regarding stereochemical integrity of Cu-allenyl intermediates. *J. Am. Chem. Soc.* **2018**, *140*, 2643–2655; (p) Isomura, M.; Petone, D. A.; Carreira, E. M. Coordination-induced stereocontrol over carbocations: asymmetric reductive deoxygenation of racemic tertiary alcohols. *J. Am. Chem. Soc.* **2019**, *141*, 4738–4748; (q) Adamson, N. J.; Jeddi, H.; Malcolmson, S. J. Preparation of chiral allenes through Pd-catalyzed intermolecular hydroamination of conjugated enynes: enantioselective synthesis enabled by catalyst design. *J. Am. Chem. Soc.* **2019**, *141*, 8574–8583; (r) Bayeh-Romero, L.; Buchwald, S. L. Copper hydride catalyzed enantioselective synthesis of axially chiral 1,3-disubstituted allenes. *J. Am. Chem. Soc.* **2019**, *141*, 13788–13794; (s) Zhu, C.; Chu, H.; Li, G.; Ma, S.; Zhang, J. *J. Am. Chem. Soc.* **2019**, *141*, 19246–19251; (t) Zheng, W.-F.; Zhang, W.; Huang, C.; Wu, P.; Qian, H.; Wang, L.; Guo, Y.-L.; Ma, S. Tetrasubstituted allenes *via* the palladium catalysed kinetic resolution of propargylic alcohols using a supporting ligand. *Nat. Catal.* **2019**, *2*, 997–1005; (u) Yang, J.; Wang, Z.; He, Z.; Li, G.; Hong, L.; Sun, W.; Wang, R. Organocatalytic enantioselective synthesis of tetrasubstituted α-amino allenoates by dearomative γ-addition of 2,3-disubstituted indoles to β,γ-alkynyl-α-imino esters. *Angew. Chem. Int. Ed.* **2020**, *59*, 642–647; (v) Wang, H.; Luo, H.; Zhang, Z.-M.; Zheng, W.-F.; Yin, Y.; Qian, H.; Zhang, J.; Ma, S. Pd-Catalyzed Enantioselective Syntheses of Trisubstituted Allenes *via* Coupling of Propargylic Benzoates with Organoboronic Acids. *J. Am. Chem. Soc.* **2020**, *142*, 9763–9771.

[6] (a) Rodriguez, O.; Morrison, H. Photosensitized racernization of an optically active allene. *Chem. Commun.* **1971**, 679; (b) Claesson, A.; Olsson, L. I. Chiral allenes are racemised by organocuprates. *J.C.S. Chem. Comm.* **1979**, 524–525; (c) Hrváth, A.; Bäckvall, J. Mild and efficient palladium(II)-catalyzed racemization of allenes. *Chem. Commun.* **2004**, 964–965; (d) Molander, G. A.; Sommers, E. M.; Baker, S. R. Palladium(0)-catalyzed synthesis of chiral ene-allenes using alkenyl trifluoroborates. *J. Org. Chem.* **2006**, *71*, 1563–1568; (e) Gandon, V.; Lemière, G.; Hours, A.; Fensterbank, L.; Malacria, M. The role of bent acyclic allene gold complexes in axis-to-center chirality transfers. *Angew. Chem. Int. Ed.* **2008**, *47*, 7534–7538; (f) Burks, H. E.; Liu, S.; Morken, J. P. Development, mechanism, and scope of the palladium-catalyzed enantioselective allene diboration. *J. Am. Chem. Soc.* **2007**, *129*, 8766–8773; (g) Butler, K. L.; Tragni, M.; Wdenhoefer, R. A. Gold(I)-catalyzed stereoconvergent, intermolecular enantioselective hydroamination of Allenes. *Angew. Chem. Int. Ed.* **2012**, *51*, 5175–5178; (h) Hilpert, L. J.; Sieger, S. V.; Haydl, A. M.; Breit, B. Palladium- and Rhodium-catalyzed dynamic kinetic resolution of racemic internal allenes towards chiral pyrazoles. *Angew. Chem. Int. Ed.* **2019**, *58*, 3378–3381.

[7] For reports on Pd-catalyzed enantioselective allenylation of malo-

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nates with 2,3-allenol derivatives, see: (a) Imada, Y.; Ueno, K.; Kutsuwa, K.; Murahashi, S. I. Palladium-catalyzed asymmetric alkylation of 2,3-Alkadienyl phosphates. Synthesis of optically active 2-(2,3-alkadienyl)malonates. *Chem. Lett.* **2002**, 140–141; (b) Trost, B. M.; Fandrick, D. R.; Dinh, D. C. Dynamic kinetic asymmetric allylic alkylations of allenes. *J. Am. Chem. Soc.* **2005**, *127*, 14186–14187; (c) Nemoto, T.; Kanematsu, M.; Tamura, S.; Hamada, Y. Palladium-catalyzed asymmetric allylic alkylation of 2,3-allenyl acetates using a chiral diaminophosphine oxide. *Adv. Synth. Catal.* **2009**, *351*, 1773–1778; (d) Li, Q.; Fu, C.; Ma, S. Catalytic asymmetric allenylation of malonates with the generation of central chirality. *Angew. Chem. Int. Ed.* **2012**, *51*, 11783–11786; (e) Dai, J.; Duan, X.; Zhou, J.; Fu, C.; Ma, S. Catalytic enantioselective simultaneous control of axial chirality and central chirality in allenes. *Chin. J. Chem.* **2018**, *36*, 387–391; (f) Song, S.; Zhou, J.; Fu, C.; Ma, S. Catalytic enantioselective construction of axial chirality in 1,3-disubstituted allenes. *Nat. Coummun.* **2019**, *10*, 507–515.

[8] For reports on the Pd-catalyzed enantioselective syntheses of allenes *via* 2,3-allenylation of malonates with 1,3-dien-2-yl derivatives, see: (a) Ogasawara, M.; Ikeda, H.; Nagano, T.; Hayashi, T. Palladium-catalyzed asymmetric synthesis of axially chiral allenes: a synergistic effect of dibenzalacetone on high enantioselectivity. *J. Am. Chem. Soc.* **2001**, *123*, 2089–2090; (b) Ogasawara, M.; Ge, Y.; Uetake, K.; Takahashi, T. Vinyl ketones to allenes: preparation of 1,3-Dien-2-yl triflates and their application in Pd-catalyzed reactions with soft nucleophiles. *Org. Lett.* **2005**, *7*, 5697–5700; (c) Wu, Z.; Berhal, F.; Zhao, M.; Zhang, Z.; Ayad, T.; Ratovelomanana-Vidal, V. Palladium-catalyzed efficient enantioselective synthesis of chiral allenes: steric and electronic effects of ligands. *ACS Catal.* **2014**, *4*, 44–48.

[9] Ogasawara, M.; Ikeda, H.; Hayashi, T. π-Allylpalladium-mediated catalytic synthesis of functionalized allenes. *Angew. Chem. Int. Ed.* **2000**, *39*, 1042–1044.

[10] Gulías, M.; Collado, A.; Trillo, B.; López, F.; Oñate, E.; Esteruelas, M. A.; Msacreñas, J. L. Ruthenium-catalyzed (2+2) intramolecular cycloaddition of allenenes. *J. Am. Chem. Soc.* **2011**, *133*, 7660–7663.

[11] Ogasawara, M.; Okada, A.; Watanabe, S.; Fan, L.; Uetake, K.; Nakajima, K.; Takahashi, T. Synthesis, structure, and reactivity of (1,2,3-η^3-Butadien-3-yl)palladium complexes. *Organometallics* **2007**, *26*, 5025–5029.

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